

PAIRS TRADING RESOURCES

Overview

Pairs trading is a market-neutral investment strategy that involves taking opposite positions (long and short) in two highly correlated assets.

The goal is to profit from the relative price movements between the two securities, assuming their historical relationship will hold over time.

If the price of one asset rises while the other falls, the trader:

- buys the underperforming asset, and
- short-sells the outperforming one, expecting their prices to revert to the historical mean.

This approach reduces exposure to overall market direction, as it focuses on the relative rather than absolute price movements of assets.

What is Pairs Trading?

Learn the basic concept, key terminologies, and strategy logic behind pairs trading.

Link: <https://blog.quantinsti.com/pairs-trading-basics/>

Correlation vs Cointegration

In pairs selection, we begin by identifying assets with correlated prices.

However, correlation alone is not sufficient — cointegration provides statistical confirmation that two price series move together in the long run.

Link: <https://hudsonthames.org/an-introduction-to-cointegration/>

Introduction to Market Neutral Strategies and Pair Selection Ideas

A detailed guide explaining different market-neutral strategies, how pairs trading fits into them, and various pair selection techniques such as:

- correlation-based methods
- cointegration-based methods
- distance and clustering methods

Link: <https://hudsonthames.org/definitive-guide-to-pairs-trading/>

Advanced Approaches

Pair Selection using Machine Learning

Explore how machine learning can enhance the process of pair identification, using models that capture

nonlinear relationships and dynamic behavior.

Link: <https://hudsonthames.org/employing-machine-learning-for-trading-pairs-selection/>

Research Papers

1. Pairs Selection using Clustering Algorithms
Link: <https://www.sciencedirect.com/science/article/pii/S037722172200769X>
 2. Pairs Trading using Deep Learning
(Recommended only if you are comfortable with deep learning concepts or if you can learn and apply)
Link: <https://www.mdpi.com/2571-9394/6/2/24>
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More Detailed:

Statistical / Classical Approaches

1. **Pairs Trading: Performance of a Relative-Value Arbitrage Rule**
Gatev, Goetzmann, Rouwenhorst (2006)
Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=141615
2. **Pairs Trading: Quantitative Methods and Analysis**
Vidyamurthy (2004)
Link: Its a book with theory and Strategy
<https://download.e-bookshelf.de/download/0000/5844/79/L-G-0000584479-0002384386.pdf>
3. **The Cointegration Alpha: Enhanced Index Tracking and Market Neutral Strategies**
Alexander & Dimitriu (2002)
Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=315619
4. **An Adaptive Kalman Filter Approach to Pairs Trading**
Elliott, van der Hoek, Malcolm (2005)
Link: https://www.researchgate.net/profile/Binh-Do/publication/267721035_A_New_Approach_to_Modeling_and_Estimation_for_Pairs_Trading/links/54c703d50cf22d626a3609ba/A-New-Approach-to-Modeling-and-Estimation-for-Pairs-Trading.pdf
5. **High-Frequency Statistical Arbitrage Using PCA**
Avellaneda & Lee (2010)
Link: https://www.tandfonline.com/doi/full/10.1080/14697680903124632?casa_token=mNui0D1K67oAAAAA%3ANxwfa7KyP6LdSEeKI4zTEZKFCDFeNWAHIZVcU6uvhjF7IhU1vqXtlmT2YSaN9Hc9H6wWsA8fH6#d1e277
6. **Statistical Arbitrage with Cointegration in the US Market**
Chen, Tan (2013)
Link: https://www.researchgate.net/publication/256043956_Selection_of_a_Portfolio_of_Pairs_Based_on_Cointegration_A_Statistical_Arbitrage_Strategy

Machine Learning-Based Approaches

1. Pair Selection using Clustering Algorithms

Han et al. (2022)

Link: <https://www.sciencedirect.com/science/article/pii/S037722172200769X>

2. Enhancing a Pairs Trading Strategy with the Application of Machine Learning

Sarmiento & Horta (2020)

Link: <https://www.sciencedirect.com/science/article/abs/pii/S0957417420303146>

3. Statistical Arbitrage Using Machine Learning

Krauss, Do (2016)

Link: [Link](#)

4. Using Random Forests and SVMs for Pair Selection and Entry Filtering

Lee, Bondarenko (2015)

Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2681265

Deep Learning-Based Approaches

1. Pairs Trading using Deep Learning

Lee, Yang, Kwon (2022)

Link: <https://www.mdpi.com/2571-9394/6/2/24>

2. Deep Reinforcement Learning Framework for Pairs Trading

Gao et al. (2020)

Link: <https://arxiv.org/pdf/2407.16103?>

3. LSTM Encoder–Decoder for Mean-Reverting Spread Forecasting

Borovkova et al. (2019)

Link: https://re.public.polimi.it/bitstream/11311/1184582/5/IRIS_EJOR_2021.pdf

Implementation Notes (for Indian Markets)

- In India, short-selling is allowed only intraday, so strategies must be adapted to intraday data (you can avoid this as well, just be known of the fact if its there in your project or not)
- Incorporate transaction costs of approximately 0.1% of the traded amount while backtesting or simulating results.
- Focus on identifying highly liquid pairs (for example, large-cap Nifty stocks) to ensure realistic execution or fine tune your paper parameters for Nifty/Sensex stocks.